WELLSLING · STEM CHECK

WellSling STEM check

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First STEM check (kit 1.9.1 upgrade) of the TRL 3 design and the three concept options; six failed details fixed under Amish's standing instruction of 2026-10-23 (WSL-DEC-001 D10 to D14, D17); independent review recorded

WellSling STEM check

Verdict: the working principle of WellSling, lifting a 600 kg animal out of a 15 m well with a frame, a braked hand winch and two webbing bands, all worked from the rim, sits within known physics, chemistry and biology. Of 48 claims, 37 are within known science and 11 are plausible, to confirm at TRL 4. No row fails on the checks made, but the dynamic load of a kicking animal, the soil and lining at real wells and the animal's tolerance of the bands are open, not proven. The check found seven details that failed. Each was fixed in the same session under Amish's standing instruction of 2026-10-23 and closed: the winch leg missed its factor of 3 once an initial bow is counted, four joints of the kit 1.7 design could not be assembled, option B's beam pieces were too heavy to carry, and (found by the independent review) the 1.5 times proof load exceeded the winch's working rating on its third layer. Eight of the plausible items belong to the chosen design (S6, S14, S15, S25, S26, S34, S36, S39) and need a winch datasheet, a site, a veterinarian or a dummy trial, so `stem_check` is **open**. Method and rules: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which reads the results of WSL-CAL-001 (`docs/04-calcs/sizing.py`) and the parametric model.

> **Safety:** WellSling lifts a heavy, frightened animal over an open well. Nothing in this check replaces a proof-load test of the real tripod, winch, rope and rigging at 1.5 times the hook load, which is TRL 4 work. Nobody goes down the well at any stage.

Claims checked

Table 1. Claims, governing science and verdicts. S rows are the chosen design (option A) and the idea as a whole; B and C rows are the other concept options (WSL-OPT-001).

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Hook load 6.49 kN for a 600 kg animal and its rigging	Physics	Mechanics and statics	Weight W = m g; the hook carries the animal, the rigging below it and the hanging rope	in 6.492 kN, out 6.492 kN	**Within known science**	WSL-CAL-001 A1, A2
S2	Wire rope safety factor 6.0 on the bare rope and 5.4 with its pressed ferrule eye	Physics	Strength of materials	Rope tension through a 97 % sheave against the 40 kN minimum breaking load of 8 mm 6 x 19 IWRC 1770 rope, times 0.9 for a ferrule-secured eye (catalogue class); 5 required (CAL assumption 5), above the 3.15 to 3.35 of ISO 4308-1 for mechanism groups M1 and M2	5.38 >= 5	**Within known science**	WSL-CAL-001 B1; independent review
S3	Sheave pitch diameter to rope diameter ratio 25	Engineering and technology	Standards practice	Bending fatigue of wire rope grows as D/d falls; 18 used as the screening minimum (CAL assumption 5)	25 >= 18	**Within known science**	WSL-CAL-001 B2
S4	Winch pull on the third rope layer covers the rope tension (87% used)	Physics	Mechanics and statics	Drum torque fixed by the gearbox: pull falls in proportion to the layer radius (rated 1,000 kg on layer 1)	0.8702 (share of capacity) <= 1 (share of capacity)	**Within known science**	WSL-CAL-001 B5
S5	At least three dead turns stay on the drum with the animal 15 m down (5.9 turns), as well	Engineering and technology	Standards practice	The rope end is held by the winch's own clamp; the dead turns take most of the tension off it by friction;	5.9 turns >= 3 turns	**Within known science**	WSL-CAL-001 B4

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	as the drum's rope clamp			three turns used as the screening minimum			
S6	Crank force at most 250 N through the lift (200 N at the landing height)	Biology	Ergonomics	Handle force scales with rope tension and layer radius from a catalogue-class 230 N at the rated load (CAL assumption 9)	200 N against the 250 N target (PR-011); rests on the handle force of the winch bought, not yet confirmed. To close: winch datasheet and a force gauge at TRL 4 (WSL-DEC-001 item 1)	**Plausible, to confirm at TRL 3**	Open (see the check)
S7	Crank work 147 kJ covers 114 kJ of lift and the gear and sheave losses	Physics	Thermodynamics	Conservation of energy, W = m g h plus friction losses	in 147.2 kJ, out 147.2 kJ	**Within known science**	WSL-CAL-001 B8, B9
S8	Whoever cranks works at about 94 W near the top for a turn of 2 to 3 minutes; in a crew of three each averages 31 W	Biology	Human physiology and biomechanics	Short bouts up to about 300 W and about 75 W sustained (kit limits); the 94 W is above the sustained figure, so turns are short	94.3 W while cranking <= 300 W while cranking	**Within known science**	WSL-CAL-001 B8; independent review
S9	A 15 m lift takes about 30 min at 15 crank turns a minute	Mathematics	Orders of magnitude	Rope speed = drum circumference / gear ratio x crank speed (mean layer radius about 58 mm)	claimed 30.14 min, estimate 32 min (ratio 0.94)	**Within known science**	WSL-CAL-001 B8
S10	Tripod does not tip with the hook line leaning 10	Physics	Mechanics and statics	Static equilibrium of the three-leg frame with the rope pull on the head; a leg unloads when its	19.4 deg >= 10 deg	**Within known science**	WSL-CAL-001 C3, C4

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	deg (tips at 19.4 deg)			compression reaches zero			
S11	Leg buckling factor 7.3 at the safe working load	Physics	Strength of materials	Euler load of a pinned 4.5 m strut, pi^2 E I / L^2; factor 3 required	7.31 >= 3	**Within known science**	WSL-CAL-001 C5
S12	Leg 1 (winch leg) stays below a third of yield in combined compression and bending with an initial bow of L/200 (kit 1.7 tube 76.1 x 3.6 at its own loads: factor 2.89; now 88.9 x 3.2: factor 3.7)	Physics	Strength of materials	Second-order elastic stress with an equivalent bow e0 = L/200 (EN 1993-1-1 Table 5.1, curve c, cold-formed hollow section); factor 3 required	83.5 MPa <= 103.3 MPa	**Within known science**	WSL-DEC-001 D10 (was Fails, fix proposed); CAL C6
S13	Spread chains stop the feet sliding: thrust is 0.59 of the foot load, above timber-on-wet-soil friction; each chain crosses 1.77 m of a 3 m well mouth near its edge, taut	Physics	Mechanics and statics	Coulomb friction 0.4 (CAL assumption 12) against the leg thrust; the chains take the difference (factor 22 on their working load limit); catenary sag w L^2 / 8 T	0.59 (thrust / vertical) >= 0.4 (thrust / vertical)	**Within known science**	WSL-CAL-001 C7, C8
S14	Soft wet ground carries the pads at 15 kPa, 22 kPa with the hook line leaning 10 deg	Physics	Mechanics and statics	Bearing pressure = foot load / pad area, against 75 kPa assumed for soft wet ground (CAL assumption 12) and 40 kPa at the low end of soft clay	15 kPa at the safe working load with the hook line vertical; 22 kPa at SWL and 33 kPa at proof with the 10 deg lean, below 40 kPa; a pad near	**Plausible, to confirm at TRL 3**	Open (see the check)

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					the shaft has a smaller bearing wedge. To close: bearing check at the first trial site, TRL 4 (WSL-DEC-001 item 4)		
S15	Pads and bearers add little to the earth pressure on the well wall (0.8 and 2.1 kPa where the load spread reaches it), and the rim and lining hold	Physics	Mechanics and statics	2:1 load spread under a footing, compared with the overburden of soil at that depth (about 18 kN/m3); the strength of an old brick or dry-stone lining is not known	Added stress at most 0.14 of the overburden; the lining that actually fails is the unknown, so the rim and lining are a go or no-go check at every well (WSL-BLD-001 safety stop S4). To close: site check at every well	**Plausible, to confirm at TRL 3**	Open (see the check)
S16	Leg pins in double shear far below grade 8.8 shear yield	Physics	Strength of materials	tau = F / (2 A); grade 8.8 shear yield about 0.6 x 640 MPa	17.65 MPa <= 128 MPa	**Within known science**	WSL-CAL-001 D1
S17	Sheave axle bending within a third of S355 yield	Physics	Strength of materials	Simply supported pin, M = F L / 4, sigma = M / (pi d^3 / 32)	95 MPa <= 118.3 MPa	**Within known science**	WSL-CAL-001 D3
S18	Spreader beam bending factor 6.3	Physics	Strength of materials	Cantilever moment from 60 % of the animal on one band; factor 3 on S355	6.301 >= 3	**Within known science**	WSL-CAL-001 E1
S19	Spreader cross bar bending factor 4.8	Physics	Strength of materials	Cantilever from the bolts to the tip lug; factor 3 on S355	4.826 >= 3	**Within known science**	WSL-CAL-001 E2
S20	Band end yoke bars	Physics	Strength of materials	Simply supported 25 mm bar, band	4.781 >= 3	**Within known science**	WSL-CAL-001 E7

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	bending factor 4.8			leg at mid span; factor 3 on S355			
S21	Bridge runner bending factor 3.4 with the animal lying at mid span	Physics	Strength of materials	Simply supported RHS over 4.0 m, 1.25 x the animal on landing, 40 % on one runner	3.381 >= 3	**Within known science**	WSL-CAL-001 F1, F2
S22	Plywood deck bending factor 3.4 under one hoof	Physics	Strength of materials	Plate strip of 300 mm over 500 mm runner centres; about 25 MPa characteristic bending strength of exterior plywood (CAL assumption 15)	3.398 >= 3	**Within known science**	WSL-CAL-001 F3
S23	Hooves clear the deck by 168 mm at the landing height, so the bridge slides under	Mathematics	Units and dimensional analysis	Stack of heights from the head sheave down to the hooves, from the model	168.5 mm >= 150 mm	**Within known science**	WSL-CAL-001 F5
S24	Average band pressure on the animal about 17 kPa	Mathematics	Orders of magnitude	Belt on a drum: p = T / (w R) for each leg tension T over band width w at wrap radius R	claimed 16.54 kPa, estimate 20.8 kPa (ratio 0.80)	**Within known science**	WSL-CAL-001 E6
S25	Two plain bands at about 17 to 21 kPa for about 30 min do not injure a 600 kg cow or buffalo	Biology	Plant and animal biology	Large-animal rescue practice favours wide padded bands under chest and flank and short suspension; no tolerable pressure or time for cattle was found in a cited source	No cited threshold; held as a question for the veterinary partner (sling width, pressure, chest and flank share, suspension time, risk of regurgitation). To close: veterinary review before any live lift, TRL	**Plausible, to confirm at TRL 3**	Open (see the check)

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					4 (WSL-DEC-001 item 5)		
S26	The passer pole, held from the rim, can push the J-head under a lying animal 15 m down (pole Euler load about 115 N)	Physics	Strength of materials	A 16 m, 40 x 2 mm aluminium pole buckles at pi^2 E I / L^2 if pushed end-on; the pole's own 17.6 kg hangs from the top hand, so the push at the head is its weight plus that	Euler load 115 N pinned over 16 m, weight about 173 N; the force needed to slide the J under an animal in mud is not known. To close: dummy trial in a test pit, TRL 4 (WSL-BLD-001 first checks)	**Plausible, to confirm at TRL 3**	Open (see the check)
S27	Passer reaches 16.5 m below the rim	Mathematics	Units and dimensional analysis	Eight 2 m sections and the J-head below the top hand	16.5 m >= 15 m	**Within known science**	WSL-CAL-001 G3
S28	Hot-dip galvanising keeps the steel parts from rusting for decades beside a well	Chemistry	Environmental and atmospheric chemistry	Zinc corrosion rate 0.1 to 0.7 um/a (C2, rural), 0.7 to 2.1 um/a (C3) and 2.1 to 4.2 um/a (C4) (ISO 9223, via AGA); mean coating 70 um on 3 to 6 mm steel and 85 um over 6 mm (EN ISO 1461:2022)	70 um at 2.1 to 4.2 um/a = 17 to 33 years beside a well with manure and splash (C3 to C4); longer in dry storage; worn surfaces (pin holes, sheave groove) lose their zinc and are inspected	**Within known science**	AGA (2026); Wedge Galvanizing (2024)
S29	Plain polyester webbing bands suit wet, muddy wells	Chemistry	Materials and polymer chemistry	Polyester is resistant to most mineral acids but is damaged by alkalis (sling user instructions to EN 1492-1)	Water and mud do not weaken polyester; lime, ash, cement slurry, manure slurry and strong alkaline cleaners can: wash in clean water, dry out of the sun, and	**Within known science**	Certex Powertex user manual (checked 2026-10-25); WSL-BLD-001 care notes

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					inspect for abrasion, cuts and hoof damage before each use		
S30	Air at the bottom of a well can be low in oxygen or hold toxic gas, so a person can collapse within seconds; working from the rim removes that exposure	Biology	Toxicology	Oxygen-deficient and toxic atmospheres in confined spaces (NIOSH FACE 85-02; CDC MMWR 1989)	Several rescuers die in a row when each goes down after the last; the kit's method needs nobody below the rim at any stage	**Within known science**	WSL-PRB-001 sources
S31	Every spread chain shackle has a hole of its own and pulls in line with its chain	Engineering and technology	Manufacturing tolerances	One pin per hole; a shackle loaded off its centre line loses working load (manufacturers' side-load rules)	Kit 1.7: one 20 mm chain eye per foot for the shackles of two chains, drawn with a vertical pin through a horizontal hole: it could not be assembled. Now two fins per foot, each square to its chain, and chains with enlarged end links	**Within known science**	WSL-DEC-001 D11 (was Fails, fix proposed)
S32	All four winch base bolts can be fitted with the bracket welded on leg 1	Engineering and technology	Manufacturing tolerances	A bolt and nut need clear space beside the tube under the bracket	Kit 1.7: the bolt pattern was centred 70 mm off the leg's centre line, so one row of holes sat 10 mm from it, over the tube. Now the bracket is 300 x	**Within known science**	WSL-DEC-001 D12 (was Fails, fix proposed)

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					300 mm and the pattern is centred 140 mm to the side: both rows clear the tube by 5 mm or more		
S33	The 240 mm band eye bears on a straight bar across its full width	Engineering and technology	Standards practice	Webbing slings wider than 75 mm should bear on an essentially straight part of the lifting attachment; the eye angle should not exceed 20 deg (sling user instructions to EN 1492-1)	Kit 1.7 shackled each 240 mm sewn eye straight onto a 1 t bow shackle with a 16 mm jaw: it could not be fitted and would have loaded the webbing at its edges. Now a band end yoke: a removable 25 mm bar 258 mm between side plates through the eye	**Within known science**	WSL-DEC-001 D13 (was Fails, fix proposed); CAL E7
S35	The guide line eyes on the spreader beam clear the end sleeves and the cross bar bolts	Engineering and technology	Manufacturing tolerances	Two parts cannot occupy the same space; a bolt head and washer need a clear seat	Kit 1.7: a 60 mm eye plate stood on the beam top 35 mm from each end, over the sleeve and M16 bolt at 500 mm. Now each eye is welded to an end cap and points along the beam	**Within known science**	WSL-DEC-001 D17 (was Fails, fix proposed)
S34	Three people set up the kit in about 20 minutes	Engineering and technology	Standards practice	Task-by-task time estimate (CAL G4)	20 min against 20 min: at the limit, and passing the bands is extra. To close: timed drill, TRL 4 (WSL-BLD-001 first checks)	**Plausible, to confirm at TRL 3**	Open (see the check)

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S36	A frightened animal kicking in the bands stays within the 1.5 times proof margin	Physics	Acoustics and vibration	Dynamic load on a hanging mass; the automatic brake allows no free fall, and the bands and rope stretch act as a spring	At a dynamic factor of 1.5 the rope is at 4.0 on its break and the steel at its proof stresses (leg 1 136 MPa, factor 2.3 on yield). A slip or a sudden catch could give more; the dynamic factor of a kicking 600 kg animal is not known. To close: load cell on the hook during the dummy trial, TRL 4	**Plausible, to confirm at TRL 3**	Open (see the check)
S37	Winch drum pitch diameter to rope diameter ratio 13.5	Engineering and technology	Standards practice	ISO 4308-1 Table 2: drum coefficient h1 at least 11.2 for mechanism group M1 and 12.5 for M2 (a rescue kit is used rarely)	13.5 >= 12.5	**Within known science**	ISO 4308-1; independent review
S38	The winch holds the kit's 1.5 times proof load on its third layer (0.69 kN m at the drum)	Engineering and technology	Standards practice	Drum torque = rope tension x layer radius; the winch is bought statically tested by its maker to 1.5 times its rated load on the first layer	0.692 kN m <= 0.795 kN m	**Within known science**	WSL-DEC-001 D18 (was Fails, fix proposed: the proof exceeded the winch's working rating on layer 3)
S39	The winch can break the animal free of mud at the bottom of the well	Physics	Fluid mechanics	Suction and adhesion of wet mud add to the weight when a body is pulled out; no figure for a 600 kg animal was found	If the crank force rises past 250 N with no movement, the crew stops and does not force it (WSL-BLD-001 safety stop S5); the breakout	**Plausible, to confirm at TRL 3**	Open (see the check)

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					force is not known. To close: dummy trial in a mud-bottomed test pit, TRL 4		
B1	Option B: gantry beam RHS 140 x 80 x 4 over 5.4 m, bending factor 3.1	Physics	Strength of materials	Simply supported beam, two hoist loads 900 mm apart (60 % and 40 % of the animal); factor 3 on S355	3.08 >= 3	**Within known science**	Open (see the check)
B2	Option B: A-frame legs SHS 60 x 3, Euler factor 39	Physics	Strength of materials	Euler load of a pinned strut; each leg carries half the end reaction along its axis	39.4 >= 3	**Within known science**	Open (see the check)
B3	Option B: hand chain pull stays within one person's reach (about 120 N per hoist at 0.4 of a 1 t rating)	Biology	Ergonomics	Hand chain pull scales with load from a catalogue-class 300 N at the rated load (assumed, not cited)	Two operators must pull together to keep the animal level; the hand chains hang over the well mouth and are pulled from the rim at an angle. Not carried forward unless option chosen	**Plausible, to confirm at TRL 3**	Open (see the check)
B4	Option B: heaviest beam piece within a two-person carry of 35 kg (two halves: 38.6 kg; three sections: 25.7 kg)	Biology	Ergonomics	Mass of steel = density x volume; 35 kg two-person carry limit (PR-013)	25.7 kg <= 35 kg	**Within known science**	WSL-DEC-001 D14, option B model (was Fails, fix proposed)
C1	Option C: the back guy (6.5 kN) is held by a staked	Physics	Mechanics and statics	Moment of the hook load about the feet line taken by the	Guy tension 6.5 kN at 29 deg; no holding capacity for a staked	**Plausible, to**	Open (see the check)

#	Claim (number and unit)	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	anchor plate in soft ground			guy's lever arm; anchor holding in soil depends on the soil and stake depth	plate in wet soil was found in a cited source. A pull test would close it if option C were chosen	**confirm at TRL 3**	
C2	Option C: a farm tractor can haul the 6.7 kN rope tension	Physics	Mechanics and statics	Drawbar pull about 0.4 x the weight of a 2 t tractor on soft ground (traction coefficient, order of magnitude)	claimed 6.693 kN, estimate 7.85 kN (ratio 0.85)	**Within known science**	Open (see the check)
C3	Option C: three turns on the snubbing post let one person hold the load with about 396 N	Physics	Mechanics and statics	Capstan equation T1 / T2 = exp(mu theta), steel rope on a steel post mu about 0.15; 400 N taken as a short two-handed pull (assumed)	396 N <= 400 N	**Within known science**	Open (see the check)
C4	Option C: a tractor can lift the animal smoothly and stop on command	Engineering and technology	Control	A tractor at creep speed raises the animal 17.6 m in about a minute; a jerk or a missed stop loads the rope and the animal suddenly	Control rests on the driver and a signaller; no brake holds the load except the post turns. Not carried forward unless option chosen	**Plausible, to confirm at TRL 3**	Open (see the check)
C5	Option C: sheerlegs CHS 88.9 x 3.2, 4.74 m, Euler factor 7.6, factor 12.0 on yield with an initial bow of L/200	Physics	Strength of materials	Euler load of a pinned strut against the leg force from the hook load, the rope fall and the guy; second-order stress with a bow of L/200 (EN 1993-1-1)	7.65 >= 3	**Within known science**	Open (see the check)

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. The kit's verdict text says "to confirm at TRL 3"; for a repo already at TRL 3, every open item here is confirmed at TRL 4, as the check column says.

Failed details fixed under the standing instruction

Each was a detail that failed while the working principle held, so the fix rule applied: the best fix for the design and its constructability was made in this session, recorded in the design decisions register as decided under Amish's standing instruction of 2026-10-23, and the check was rerun so the row closed. Amish can reverse any of them.

Table 2. Fixes.

Row	What failed	Fix	Record
S12	Leg 1 carries the rope pull 121 mm off its axis. With the initial bow of L/200 that EN 1993-1-1 gives for a cold-formed hollow section, the 76.1 x 3.6 mm tube reaches 104 MPa at the safe working load, a factor of 2.98 against the 3 required (2.89 with the secant amplification adopted after the independent review). The kit 1.7 note checked buckling with the Euler load only	All three legs 88.9 x 3.2 mm: factor 3.7 (83 MPa); 1.9 kg more per leg, 31.8 kg each, still under the 35 kg carry limit	WSL-DEC-001 D10
S31	Each foot had one 20 mm chain eye for the shackles of two spread chains, drawn with a vertical pin through a horizontal hole	Two chain eye fins per foot, each square to its own chain, and chains with enlarged end links to take the shackle bows	WSL-DEC-001 D11
S32	The winch bolt pattern was centred 70 mm off leg 1's centre line, so one row of holes sat over the tube and could not take a nut	Bracket 300 x 300 mm, centred 110 mm to the side; bolt pattern centred 140 mm to the side; both rows clear the tube	WSL-DEC-001 D12
S33	Each 240 mm band eye was shackled straight onto a 1 t bow shackle with a 16 mm jaw. It cannot be fitted, and a webbing sling wider than 75 mm must bear on an essentially straight part	Band end yoke: a removable 25 mm bar through the eye, 258 mm between side plates, with a lug for the shackle; adjusting chain in use 200 mm so the hooves still clear the deck (169 mm)	WSL-DEC-001 D13
S35	The spreader's guide line eyes stood on the beam top 35 mm from each end, over the sleeve and M16 bolt at 500 mm	Each eye welded to an end cap, pointing along the beam	WSL-DEC-001 D17
S38	Independent review: the kit's proof load of 1.5 times the hook load puts 10.0 kN on the rope, 130 % of the winch's working rating on its third layer	The winch is bought statically tested by its maker to 1.5 times its rated load, and the proof load is held on the brake: 0.69 kN m at the drum against the 0.80 kN m of that test	WSL-DEC-001 D18
B4	Option B's 5.8 m gantry beam in two halves weighed 38.6 kg a piece, over the 35 kg two-person carry	Three sections of about 1.93 m (25.7 kg) with two bolted splices	WSL-DEC-001 D14

The fixes add 13.3 kg of yokes and about 1.4 kg of other steel below the hook: the design hook load rises from 648 kg to 662 kg (6.49 kN), the rope factor falls from 6.1 to 6.0 (need 5), the winch runs at 87 % of its capacity on the third layer and the crank force at the landing height is 200 N (need at most 250 N). Every steel part keeps a factor of 3 or more (WSL-CAL-001 v0.2).

Fields not relevant

All five fields are touched: physics (statics, strength of materials, thermodynamics of the lift), chemistry (zinc coating and polyester webbing), biology (the bad air the kit avoids, the animal in the bands, human effort and carrying), mathematics (energy, time, heights and reach) and engineering and technology (rope practice, joints that must assemble, setup time, control of a draught haul). None is marked not relevant.

Concept options

- **Option A, tripod and hand winch (the design since kit 1.7):** within known science. Eight rows are plausible and open for TRL 4 (S6 winch handle force, S14 soil bearing, S15 rim and lining, S25 animal tolerance of the bands, S26 passer push force, S34 setup time, S36 dynamic load of a kicking animal, S39 breaking free of mud). Six failed details are fixed (S12, S31, S32, S33, S35, S38).
- **Option B, portable gantry and two chain hoists:** within known science. The beam and A-frames have factors of 3.1 and 39 (B1, B2). The hand chain pull is plausible (B3). The beam was resectioned to meet the carry limit (B4).
- **Option C, sheerlegs and draught haul:** within known science for the frame (C5, factor 7.6 on Euler and 12.0 on yield with an initial bow), the tractor pull (C2) and the snubbing post (C3, about 400 N to hold with three turns). The guy anchor's holding in soft ground (C1) and smooth control of a tractor haul (C4) are plausible only and carry the option's main risk.

No option has a **Fails** row, so none is dropped.

Independent review

A second agent, run headless on 2026-10-25 with only four files (this check as it stood, WSL-CAL-001 v0.2, WSL-PRS-001 and `sizing.py`) and none of the author's reasoning, tried to refute the headline results and every "Within known science" row. It reproduced the arithmetic of almost every row by hand. Its findings and what changed:

Table 3. Independent review findings and responses.

Finding	Reviewer's verdict	Response
No dynamic or snatch load case: a kicking animal could raise the loads well past static	Missing check	Added S36 (plausible, open): the 1.5 times proof covers a dynamic factor of 1.5 with the parts at their proof stresses; a load cell on the hook in the TRL 4 dummy trial will measure it. The automatic brake allows no free fall
The 1.5 times proof puts the winch at 130 % of its working rating on the third layer	Weakened	Failed detail fixed under the fix rule: winch bought statically tested to 1.5 times its rating, proof held on the brake (S38, WSL-DEC-001 D18)
Rope factor ignores the end termination; drum D/d of about 12.5 is below the 18 screening value	Weakened	S2 now 5.4 with a 90 % ferrule eye (need 5); S37 added: drum pitch D/d 13.5 against 12.5 for ISO 4308-1 group M2 (the 18 is the sheave's screening value)
Leg 1 amplification should be the secant of a constant moment, 1.20 not 1.16	Weakened	Adopted: 83 MPa, factor 3.7 (C6, S12)
Pad pressure checked only with a vertical hook line; 75 kPa is the top of the soft-clay range	Weakened	C10 added: 22 kPa at SWL and 33 kPa at proof with the 10 deg lean, below 40 kPa (S14, still open for a site check)
The rim check compares stress with overburden and says nothing about the lining that fails	Missing check	S15 changed to plausible: the rim and lining are a go or no-go check at every well (safety stop S4 rewritten)
Lift is 18.2 m of rope wound in, not 17.6 m; time about 30 min	Weakened	Adopted (B8, B9, S9)
Crank power compared as a crew average	Weakened	S8 now compares the working person (94 W) with short-bout capacity and states 2 to 3 minute turns (safety stop S5)
Spread chains cross the well mouth	Missing check	C11 added: each chain runs 1.21 m from the axis and crosses 1.77 m of a 3 m well near its edge, taut (34 mm sag), clear of the animal; harness to step over it (S4)
Winch bracket, welds and bolts not checked; 5 mm clearance cannot fit a nut	Missing check	D4 added (bolts 1.4 kN tension, welds 6 MPa). The 5 mm is between the nut's corner and the tube beside it; the nut sits beside the tube, not over it
Single-lug spreader cannot hold a 60/40 share without pitching	Weakened	E1 now says the shackle moves to the lug hole 50 or 100 mm toward the heavier band; that is what the five holes are for
Plywood strength taken as characteristic, not design	Weakened	F3 now notes the glued cleat under the same line carries most of the hoof load; bare plywood with wet factors would be near 2
Landing clearance has no tolerance stack	Weakened	F5 now notes that the landing height is a winch position with 500 mm of reserve before the rope fall reaches its minimum
Galvanising life should use C4 beside manure; add manure slurry for polyester	Weakened	S28 and S29 updated (17 to 33 years; manure slurry, abrasion and hoof cuts)
Setup time has no margin	Weakened	PR-006 now at risk; S34 stays open

Finding	Reviewer's verdict	Response
Mud breakout not considered	Missing check	Added S39 (plausible, open) and safety stop S5: stop if the crank gets much harder with no movement
Option C has no bow check	Weakened	C5 now includes it: factor 12.0 on yield
Pin bending and retention; bend radius of webbing on the 25 mm bar; runner drop case	Missing check	Pin bending 67 MPa added to D1 (R-clips on lanyards already hold the pins); the band works at 2 % of its rating on a straight bar, so the bend radius does not govern; the animal is lowered on the brake, never dropped
600 kg is not the top weight of an adult buffalo or Holstein cow	Scope question	Not changed: the 600 kg rating was decided by Amish on 2026-10-03 (WSL-DDR-001, D7). Raised as a question for Amish in docs/REVIEW.md

The reviewer upheld the bad air row (S30), the assembly fixes (S31, S33), the tripod statics and buckling (S10, S11), the spreader, cross bar and yoke bars (S18 to S20) and the energy balance (S7). The three findings it ranked highest were the missing dynamic case, the rope and winch chain (termination, drum, proof) and the ground, rim and biology unknowns; the first two are answered above and the third stays open for TRL 4.

Sources

- American Galvanizers Association. "HDG Corrosion Rates for ISO Categories C1-C5/X." Accessed October 25, 2026. https://galvanizeit.org/knowledgebase/article/hdg-corrosion-rates-for-iso-categories-c1-c5-x.
- Certex. *POWERTEX Round Slings / Webbing Slings, User Manual*. Accessed October 25, 2026. https://www.certex.dk/sites/Powertex/New%20manuals/Slings/Powertex-Slings-PRS-PWE-PWS-User-Manual-ML-20251201.pdf.
- Centers for Disease Control and Prevention. *Morbidity and Mortality Weekly Report*, 1989, as cited in WSL-PRB-001. https://www.cdc.gov/mmwr/preview/mmwrhtml/00001448.htm.
- European Committee for Standardization. EN 1993-1-1, *Eurocode 3: Design of Steel Structures, Part 1-1*, Table 5.1 (equivalent initial bow imperfections).
- International Organization for Standardization. ISO 4308-1:2003, *Cranes and Lifting Appliances: Selection of Wire Ropes, Part 1: General*, Tables 1 and 2. Accessed October 25, 2026. https://dgcrane.com/wp-content/uploads/2023/09/ISO-04308-1-2003.pdf.
- National Institute for Occupational Safety and Health. *FACE 85-02*. https://stacks.cdc.gov/view/cdc/164612.
- Wedge Galvanizing. "Hot Dip Galvanizing to EN ISO 1461:2022." 2024. Accessed October 25, 2026. https://www.wedge-galv.co.uk/wp-content/uploads/2024/10/4b-HDG-to-EN-1461-2024.pdf.
- WellSling sizing calculations, WSL-CAL-001 v0.2, and the parametric model `cad/src/model.py`.